Computer Networks/0705210503

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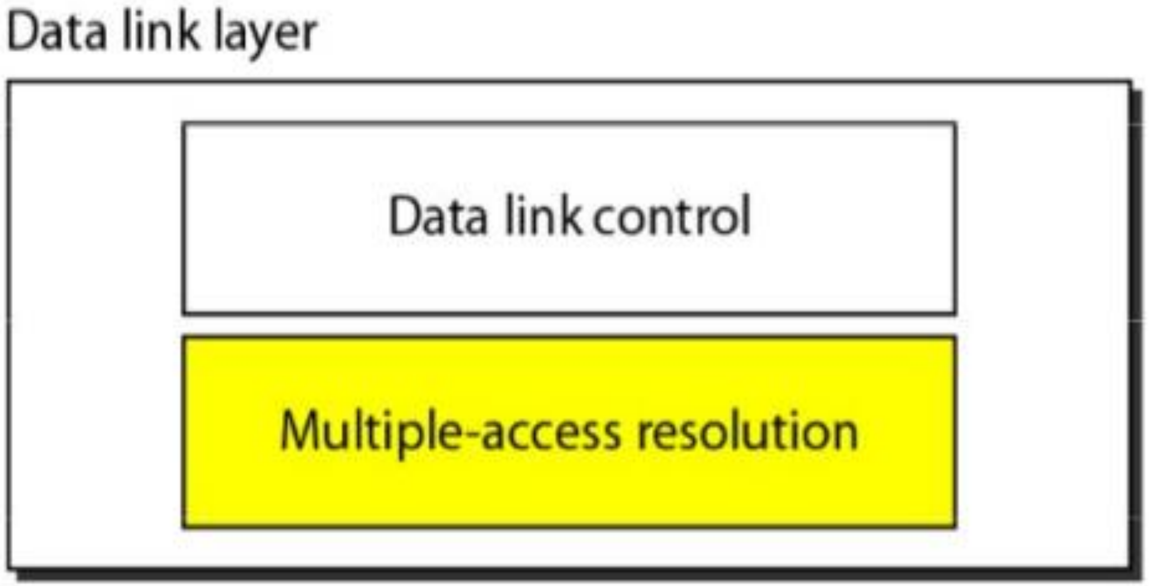
The data link layer is divided into two sublayers:

- **Logical link control (LLC) layer:** The upper sublayer is responsible for data link control, i.e., for flow and error control.
- **Media access control (MAC) layer:** The lower sublayer is responsible for resolving access to the shared media.



RANDOM ACCESS PROTOCOLS

- No station is superior to another station, and none is assigned control over another.
- No station permits, or does not permit, another station to send.
- At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.
- Each station can transmit when it desires on the condition that it follows the predefined procedure, including the testing of the state of the medium.

Two features give this method its name:

1. First, there is no scheduled time for a station to transmit. Transmission is random among the stations. That is why these methods are called random access.
2. Second, no rules specify which station should send next. Stations compete with one another to access the medium. That is why these methods are also called contention methods

ALOHA

* ALOHA is for coordinating & accessing to share communication in network channel.

* It is the simplest collision resolution protocol .

TYPES ALOHA

1. Pure Aloha:

* This is the original version of Aloha .

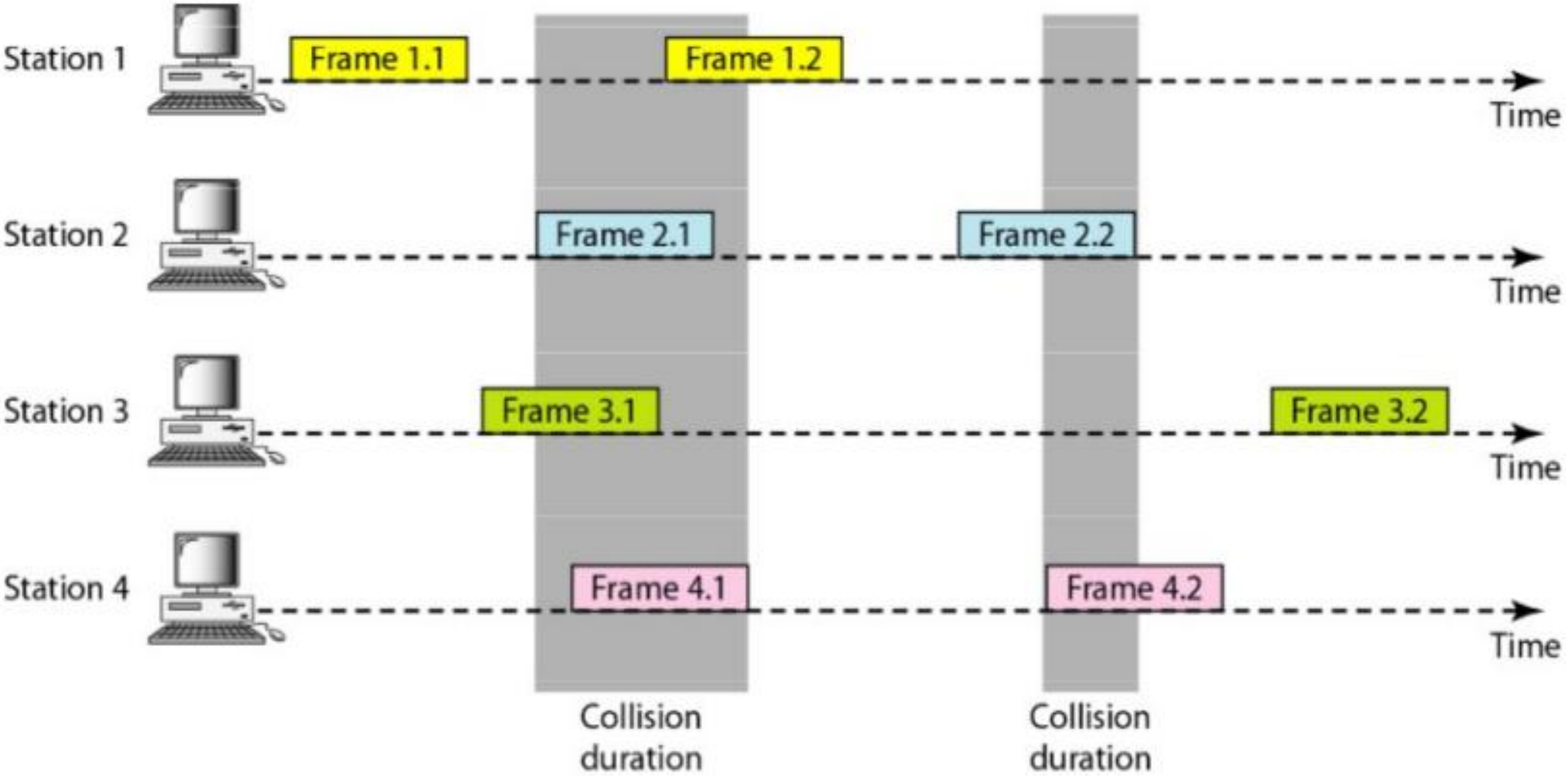
2. Slotted Aloha:

* This is the modified version of pure Aloha.

PURE ALOHA

- The original ALOHA protocol is called pure ALOHA.
- The idea is that each station sends a frame whenever it has a frame to send.
- However, since there is only one channel to share, there is the possibility of collision between frames from different stations.
- A collision involves two or more stations.
- If all these stations try to resend their frames after the time-out, the frames will collide again.
- Pure ALOHA dictates that when the time-out period passes, each station waits a random amount of time before resending its frame. The randomness will help avoid more collisions. We call this time the back-off time T_B.

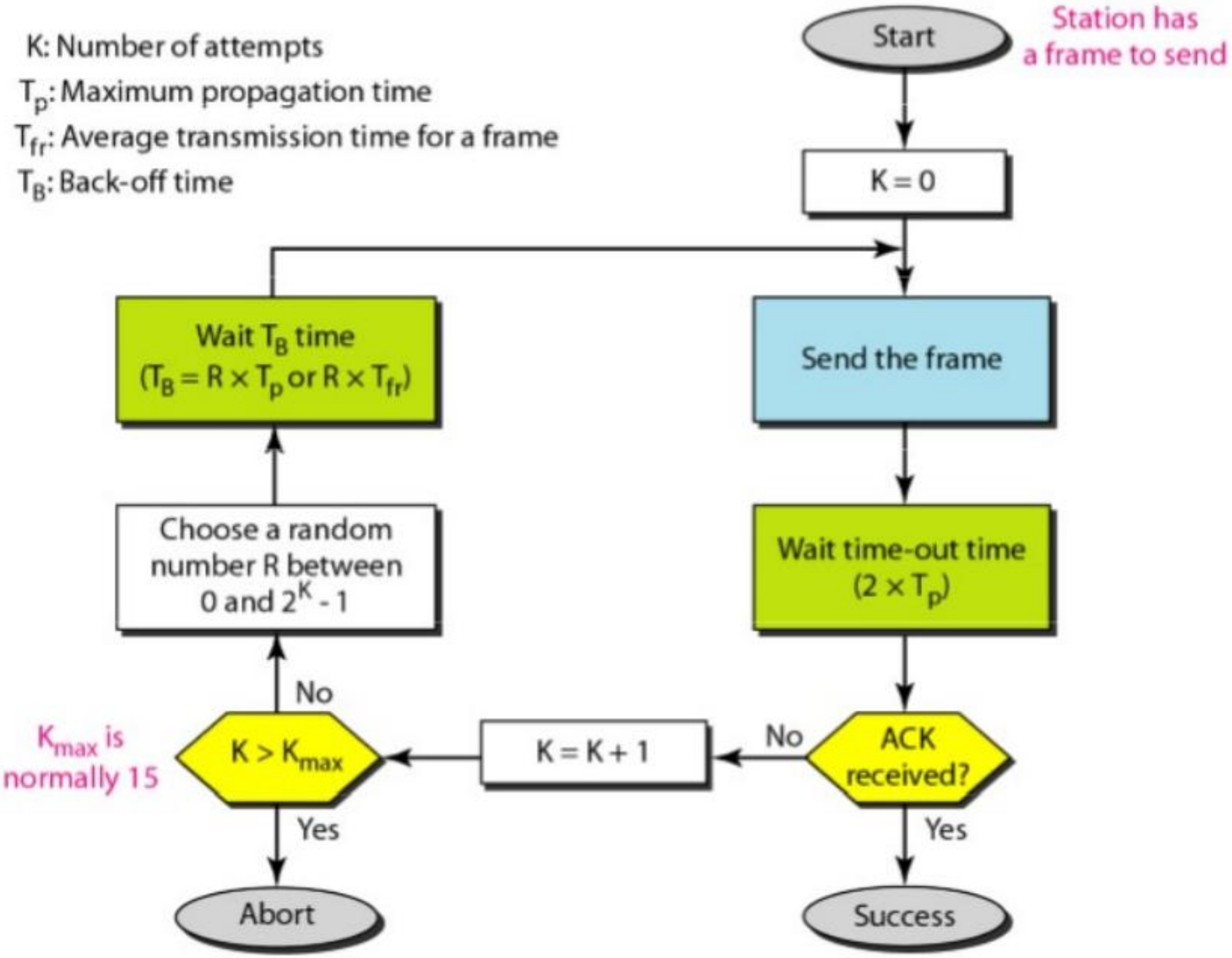
PURE ALOHA


Station 1
Frame 1.1
Frame 1.2
Time
Station 2
Frame 2.1
Frame 2.2
Time
Station 3
Frame 3.1
Frame 3.2
Time
Station 4
Frame 4.1
Frame 4.2
Time
Collision duration
Collision duration

PURE ALOHA

- The time-out period is equal to the maximum possible round-trip propagation delay, which is twice the amount of time required to send a frame between the two most widely separated stations (2 x T_p).
- The back-off time T_B is a random value that normally depends on K (the number of attempted unsuccessful transmissions).
- One common formula for calculating T_B is the binary exponential back-off.
- In this method, for each retransmission, a multiplier in the range 0 to 2^K- 1 is randomly chosen and multiplied by Tp (maximum propagation time) to find T_B.
- The range of the random numbers increases after each collision.
- The value of K_{max} is usually chosen as 15.

PURE ALOHA


K: Number of attempts
T_p: Maximum propagation time
T_{fr}: Average transmission time for a frame
T_B: Back-off time
Start
Station has a frame to send
K = 0
Send the frame
Wait time-out time ($2 \times T_p$)
ACK received?
No
Yes
Success
K = K + 1
$K > K_{max}$
K_{max} is normally 15
No
Yes
Abort
Choose a random number R between 0 and $2^K - 1$
Wait T_B time ($T_B = R \times T_p$ or $R \times T_{fr}$)

Examples

The stations on a wireless ALOHA networks are a maximum of 600 km apart. If we assume that signals propagate at 3 x 10^8 ms, we find Tp = (600 x 10^5) / (3 x 10^8) = 2 ms. Now we can find the value of T_B for different values of K.

a. For K = 1, the range is {0,1}. The station needs to generate a random number with a value of 0 or 1. This means that T_B is either 0 ms (0 x 2) or 2 ms (1 x 2), based on the outcome of the random variable.

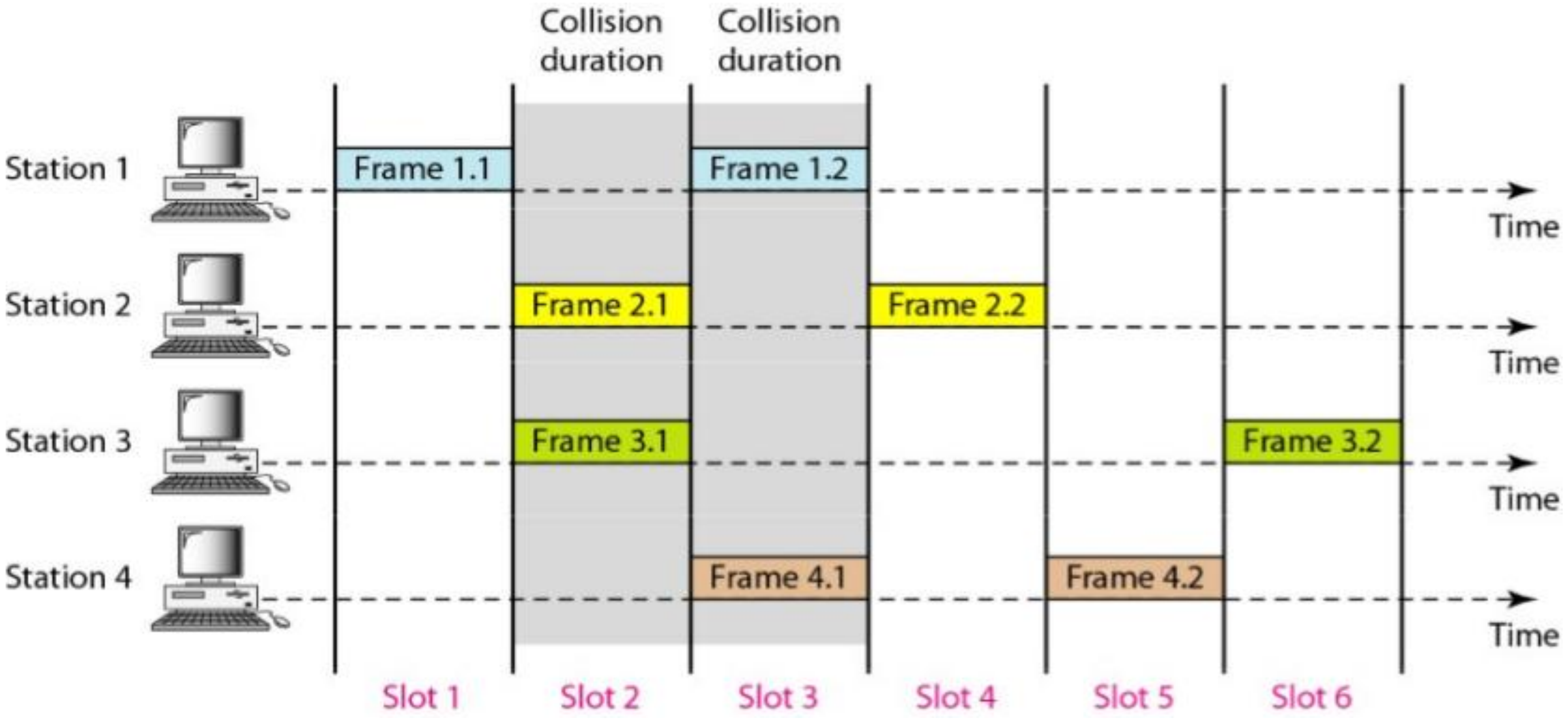
b. For K =2, the range is {0, 1, 2, 3}. This means that T_B can be 0, 2, 4, or 6 ms, based on the outcome of the random variable.

c. For K =3, the range is {0,1,2,3,4,5,6,7}. This means that T_B can be 0,2,4, ... , 14 ms, based on the outcome of the random variable.

Slotted ALOHA

- Slotted ALOHA was invented to improve the efficiency of pure ALOHA.
- In slotted ALOHA we divide the time into slots of $\mathbf{T_{fr}}$ **s** (time to send each fixed size frame) and force the station to send only at the beginning of the time slot.
- Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next time slot.
- This means that the station that started at the beginning of this slot has already finished sending its frame.
- There is still the possibility of collision if two stations try to send at the beginning of the same time slot.

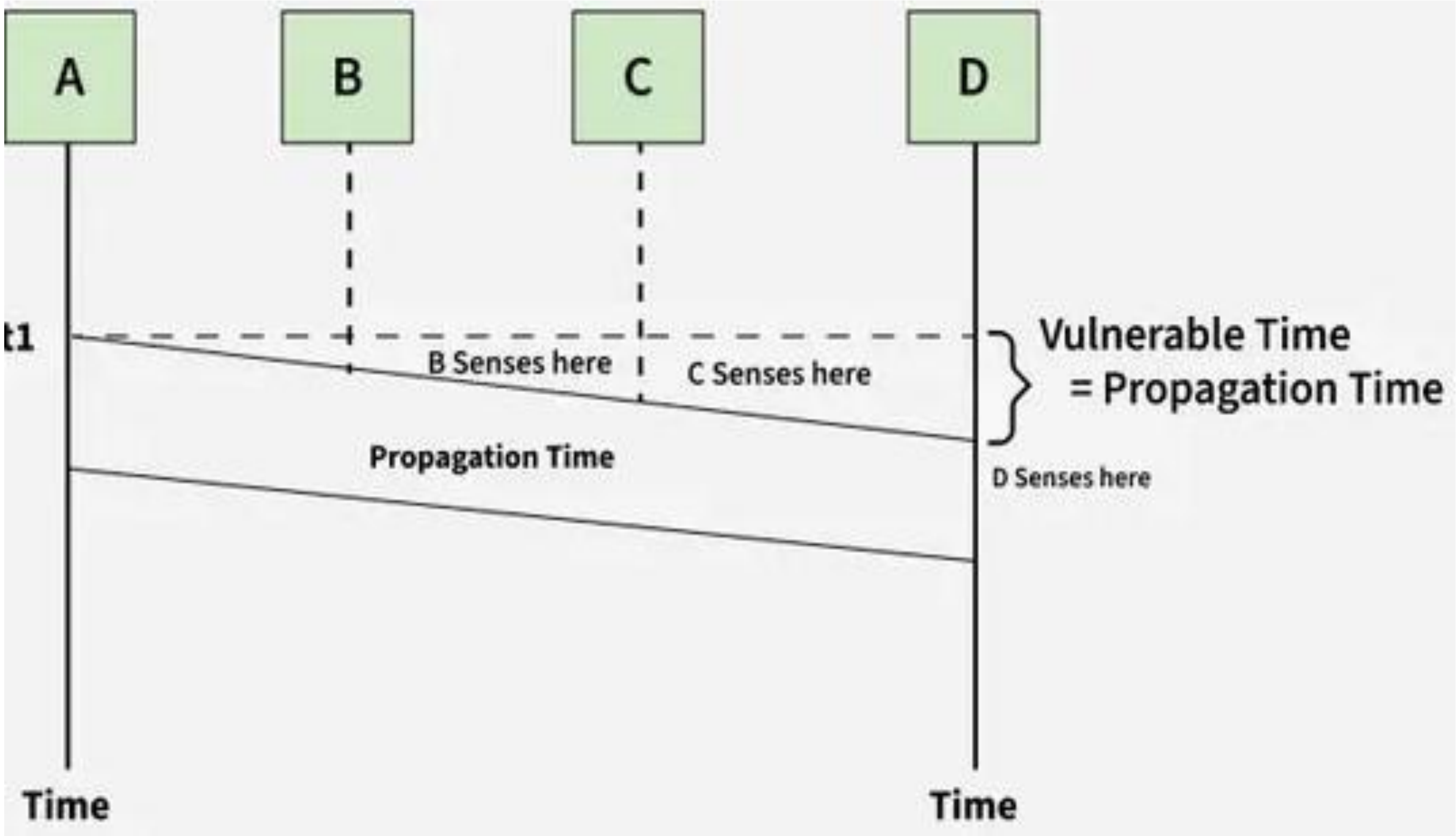
Slotted ALOHA



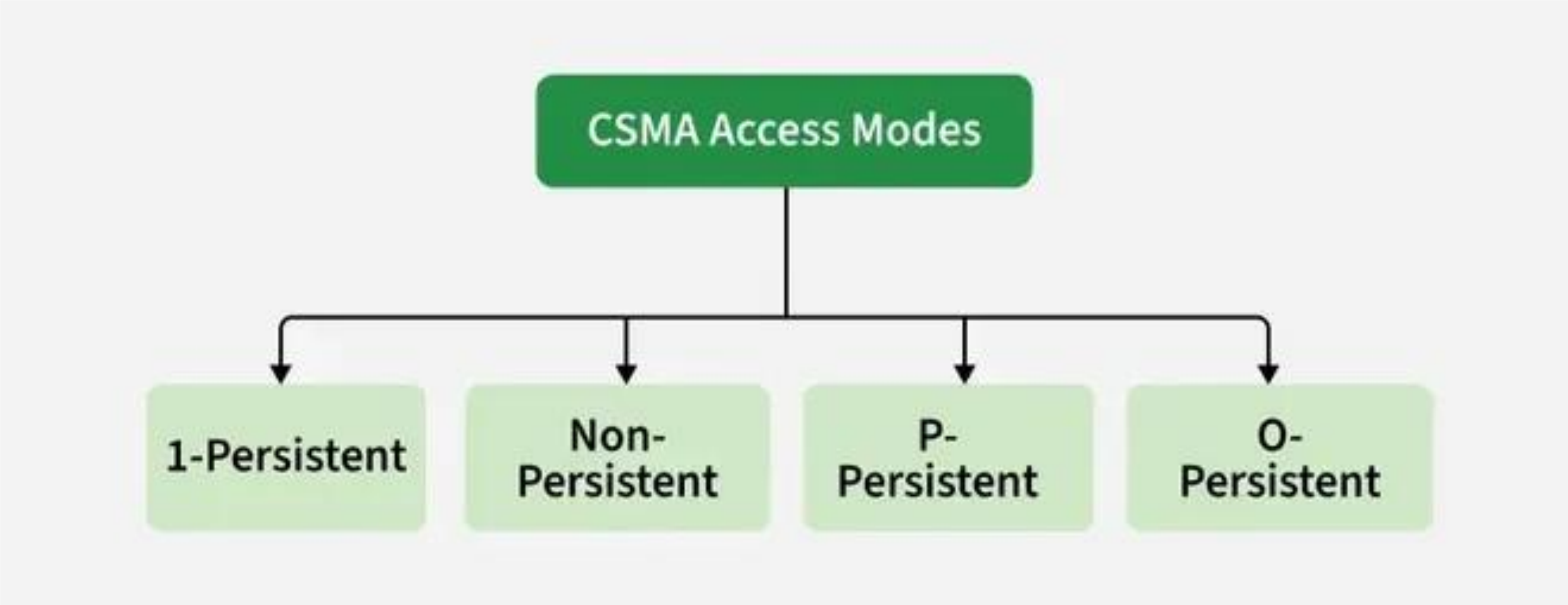
Carrier Sense Multiple Access (CSMA)

- Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data.
- If it is idle then it sends data, otherwise it waits till the channel becomes idle.
- However there is still chance of collision in CSMA due to propagation delay.
- For example, if station A wants to send data, it will first sense the medium. If it finds the channel idle, it will start sending data.
- However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data.
- This will result in collision of data from station A and B.

Carrier Sense Multiple Access (CSMA)



CSMA Access Modes:



CSMA Access Modes:

- **1-Persistent:** The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being idle and transmits unconditionally(with 1 probability) as soon as the channel gets idle.
- **Non-Persistent:** The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.
- **P-Persistent:** The node senses the medium, if idle it sends the data with p probability. If the data is not transmitted ((1-p) probability) then it waits for some time and checks the medium again, now if it is found idle then it send with p probability. This repeat continues until the frame is sent. It is used in Wifi and packet radio systems.
- **O-Persistent:** Superiority of nodes is decided beforehand and transmission occurs in that order. If the medium is idle, node waits for its time slot to send data.